SKYFORCE Interceptor: Guide Decision Record

Team 08, Anti-Drone Interceptor, Senior Track. Prepared 10 October 2026. Answers every question raised in the Guide Discussion Brief, using the latest CSV export, the sizing document, and the two prior resolution notes as the source of truth.

Each item below is marked Approved, Corrected, or Out of scope — confirmed. Nothing here invents new performance data; where a question cannot be answered without data that does not exist yet, this record says so and states exactly what export or test would be needed.

Decision record (summary)

#	Item	Decision
1	Mission energy	Adopt the latest-CSV estimates (41.44 Wh worst no-tow, 103.53 Wh tow) as the current figures. A continuous launch-to-landing run is recommended, not required, before final submission.
2	Closing speed	Line-of-sight range-rate definition approved. Cannot be independently verified with the current export; expanded CSV fields required before the numbers can be called verified.
3	Propulsion	Sensitivity table (15 to 30%) approved as sufficient evidence. Carry the 30% case forward as the headline margin. No speed-dependent model; 4.5 m/s² cap unchanged.
4	Mass / CAD	3.22 kg allowance accepted provisionally for CAD freeze. Propeller and ESC masses must be confirmed from datasheets before the literal final freeze.
5	Scope limits	Capture/recovery and site boundaries confirmed explicitly out of scope / illustrative. CAD may freeze without capture-dynamics validation.

1. Mission-energy consistency

All four arithmetic questions in this section were checked directly.

Case	Latest CSV	Buffer	Complete-mission estimate	Ceiling	Utilisation
Attack, no tow	29.85 Wh	9.7 Wh	39.55 Wh	207 Wh	19%
Defense, no tow (worst)	31.74 Wh	9.7 Wh	41.44 Wh	207 Wh	20%
Tow required	84.73 Wh	18.8 Wh	103.53 Wh	253 Wh	41%

Decision: replace the older 46.5 / 114.4 Wh figures with 41.44 Wh (worst no-tow) and 103.53 Wh (tow) as the current complete-mission estimates. Both stay well inside their ceilings. A single continuous launch-to-landing energy integration is the correct long-term fix — it would end the repeated resync problem this project has hit three times now (first the 236 Wh baseline, then 46.5/114.4 Wh, now this) — but is not required to finish the report if time is short. If it is not built, every citation of these numbers must keep the word “estimate.”

On the 18.8 vs 18.7 Wh buffer: both are arithmetically defensible and the 0.1 Wh gap has no material effect. 18.8 Wh is the more correct figure, because it comes from summing the full-precision component values (5.014 + 12.944 + 0.800 = 18.758, which rounds to 18.8) before rounding once at the end. 18.7 Wh comes from summing the already-rounded display values (5.0 + 12.9 + 0.8), which double-rounds and loses a little precision. Keep 18.8 Wh as the authoritative figure and add a footnote explaining the rounding if the displayed components are shown.

On the 12.9 Wh loaded-descent figure: this is correct, and the guide brief’s own linear-scaling check (giving 11.3 Wh) uses the less accurate relation. Hover-induced power scales with weight to the 1.5 power, not linearly, because induced velocity scales with the square root of thrust (actuator disk theory: P = T × v■, and v■ ∝ √T, so P ∝ T^1.5). This W^1.5 law is the same relation already used consistently throughout the sizing document’s tow-phase calculation (1.56 kW × (23.5/18.0)^1.5 = 2.33 kW), so 12.9 Wh is the figure to keep.

On the 253 Wh derivation: confirmed. 207 Wh comes from 172.3 Wh of modeled flight energy × 1.20 (206.8, rounds to 207). The tow addition is 38.7 Wh × 1.20 = 46.4 Wh, added to 206.8 for 253.2 (rounds to 253). The 20% margin is applied separately to the base mission and to the tow addition, not once to the combined total — this should be stated explicitly in the report as the chosen convention.

2. Closing-speed definition and audit

The proposed definition is correct and is approved as written: closing speed is the negative of the range-rate, V_close = – r■ · (v_target – v_interceptor), the line-of-sight component of relative velocity, not the magnitude of the relative velocity vector. This is the standard definition used in intercept guidance and should be documented in the code as such.

Scenario geometry check, against the exported target data:

- Attack: target speed 14.87 m/s, below the 25 m/s interceptor cap — consistent with attack (tail-chase) mode, which is selected precisely when the target is slower.
- Defense: target speed 27.00 m/s, above the cap — consistent with defense (lead-point) mode, which triggers only when the target is faster than the interceptor.
- Tow required: target speed 12.00 m/s (attack-mode geometry) — consistent, since towing is triggered by the capture point's location relative to the safe area, not by target speed.
- Abort: target speed 20.00 m/s, heading 180°, still below the cap — consistent with the reported abort cause (insufficient time before the boundary, not an unreachable target speed).

All four scenarios are geometrically consistent with their intended mode logic. Approved.

Independent verification attempted: with only target velocity and the final scalar closing speed exported, the reported values cannot be independently recomputed. A partial plausibility check is possible for the attack case: a closing speed of 25.87 m/s exceeding the interceptor's own 25 m/s cap requires the target to have a velocity component toward the interceptor along the line of sight (roughly 0.87 m/s or more) — geometrically possible for a target at a 19.65° heading relative to the engagement line, but this cannot be confirmed without the interceptor's own position and velocity at the same sample. The requested export fields (interceptor position and velocity, LOS unit vector, range rate) are required before these numbers can be called verified rather than plausible.

Decision: approve the definition; require the expanded export before the next report draft; add one additional scenario with a deliberately offset initial heading (not already on a near-collision course) specifically to show PN applying a visible correction, since the current three scenarios all start from a pre-positioned, near-collision geometry and cannot by themselves demonstrate the guidance law's corrective authority.

On whether an accepted closing-speed envelope exists for the capture mechanism: no such envelope exists in this project, and none should be invented. The point-mass model has no basis to approve or reject a given closing speed against net-capture success — this stays an explicit, named scope limitation (see Section 5).

3. Forward-flight propulsion margin

Decision: the 15 to 30% de-rating sensitivity table is sufficient for a hackathon-level report, provided every citation of it keeps the label "assumed sensitivity range, not measured data." Carry the 30% (conservative) case as the headline number in the report — margin of 1.59x at 18 kg and 1.25x at 23.5 kg, both above 1.0.

A speed-dependent thrust model (blade-element or momentum theory with advance-ratio correction) is not recommended for the remaining time budget. There is no measured thrust-polar data for the G28x9.2 CF propeller to calibrate such a model against, so building one would add assumption on top of assumption without adding real credibility — the sensitivity table already bounds the risk honestly and more cheaply.

The 4.5 m/s² commanded-acceleration cap remains unchanged. It sits far below the theoretical static-available acceleration at both masses (21.8 m/s² at 18 kg, 15.4 m/s² at 23.5 kg), so it is correctly understood as a deliberate control-smoothness and structural-load limit, not an unvalidated performance ceiling.

4. Mass ledger and CAD freeze

Decision: accept the 3.22 kg integration/growth allowance provisionally for CAD geometry freeze (arm length, span, frame layout can proceed now). Before the literal final freeze, specifically close out the two components already flagged twice in this project as unverified: the propeller mass (G28x9.2 CF) and the ESC mass (Alpha 60A), both carried over as placeholders from an earlier baseline and never checked against a supplier datasheet. These two are prioritised because they are the most likely to materially shift the ledger, unlike the remaining assumed items (frame,

fasteners, wiring), which are reasonable placeholder estimates for a scratch-built frame at this stage.

Authoritative configuration for mounting, arm layout, span and centre-of-gravity work: T-Motor U8 Lite KV100 (238 g each, published), T-Motor G28x9.2 CF 28-inch propeller, T-Motor Alpha 60A 12S ESC, 12S 8000 mAh battery (assumed 1.97 kg from a generic 180 Wh/kg LiPo density — flag this for replacement with the actual chosen battery product's datasheet mass once selected, since 180 Wh/kg is a category average, not a specific part number).

The net launcher, net and parachute assembly should also be rechecked before freezing the payload bay specifically — not primarily for mass (which is a small fraction of MTOW) but for physical envelope, since the net canister and CO2 cartridge have fixed off-the-shelf form factors that must physically fit the bay CAD is about to lock in.

5. Capture/recovery and scenario-boundary scope

5.1 Capture and recovery

Decision: only geometric intercept and sizing-level analysis is expected for this deliverable. The problem statement asks for a design report with sizing calculations, CAD, analysis/simulation results, a budget and a cost estimate within a 2-day hackathon — a deployment-dynamics model or physical net/parachute testing is outside that scope and outside the time available, and attempting either now would not add credible evidence.

Approved language: "geometric capture candidate." The phrases "successful capture" or "capture achieved" should not appear anywhere in the report for simulation-derived results, since no capture probability has been or can be established by this model.

CAD may freeze while capture/recovery performance remains explicitly out of scope, provided the net launcher, net and parachute's physical mass and envelope (not their deployment performance) are confirmed against the mass ledger and payload bay, per Section 4.

5.2 Keep-out and safe areas

Decision: the illustrative 300 m and 120 m radii are sufficient for the report as currently scoped; nominating a specific real or representative site is not required. As a low-effort upgrade, naming a representative site class in the report (for example, "consistent with DGCA's 3 km minimum zone for non-major civil airports", already cited in the prior resolution note) adds credibility without requiring actual survey data — optional, not required.

Site boundaries should remain configurable scenario parameters in the ground-control software, not CAD constraints. Confirmed: they have no bearing on the airframe design.

Minimum CSV export fields for the next run

Approved as the required schema for all future scenario exports, so closing-speed and guidance claims can be independently checked going forward:

- Time, scenario ID, and event index for the exact sample used
- Interceptor position (x, y) and velocity (Vx, Vy) at that sample
- Target position (x, y) and velocity (Vx, Vy) at that sample
- Range, line-of-sight unit vector, range rate, and the resulting closing speed
- Commanded and applied (post-limit) acceleration, and which cap was active
- Phase label and an electrical power estimate for each phase, to support a future continuous energy integration

Outcome of this record

All five decision items requested in the brief are resolved above. Two items (closing-speed independent verification, and the continuous launch-to-landing energy model) require the next simulation export before they can move from "definition approved" to "independently verified" — everything else in this brief is closed as of this record. Carry this document, the sizing calculations PDF, the architecture PDF, and the prior resolution PDF forward together as the full current source of truth.